

User Study – Manual Python Workflow (Method 1 Only)

Instructions: Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: _____ Your assignment: ☐ Method 1 (Manual Python)

Use case(s) you worked on: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

Section A – Background (answer once)

A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☒ Bachelor's ☐ Master's ☐ Doctorate ☐ Other:

A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☒ Other:

Master Student

A3. Self-rated expertise (*tick one per row*)

	Beginner	Intermediate	Advanced	Expert
Python programming	☒	☐	☐	☐
Time-series / ML	☒	☐	☐	☐
Domain knowledge (your use-case)	☒	☐	☐	☐

A4. Prior experience (*used ML frameworks/libraries before?*) ☒ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

B1. Time & Effort

Record your own times. Estimate if needed. “Extra learning time” = time spent reading docs/tutorials.

B1. Time spent (minutes)

	Method 1 (Manual Python)
Writing/testing code (end-to-end)	<u>344</u>
Extra learning time	<u>≥ 180</u>

B2. Lines of code (LoC) (*exclude blanks/comments*)

Manual Python LoC: _____

B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

B3. Were you able to finish the use-case with Manual Python?

Method 1 (Manual Python): ☐ None ☐ Partial ☒ Complete

B4. Ease by workflow step (Manual Python) (1=Very easy, 5=Very difficult; tick one per cell)

Step	1	2	3	4	5
Whole workflow	☐	☐	☐	☒	☐
Data preprocessing	☐	☐	☐	☐	☒
Feature engineering (lags/rolling)	☐	☐	☐	☒	☐
Model training	☐	☒	☐	☐	☐
Model evaluation/plots	☐	☒	☐	☐	☐

B1. Which steps did you complete? (check all that apply)

- ☒ Preprocessing
- ☒ Model configuration
- ☒ Training
- ☒ Evaluation
- ☒ Plotting/Diagnostics

B2. How difficult was each step? (1=Very easy, 5=Very difficult; tick one per row)

Step	1	2	3	4	5
Preprocessing	☐	☐	☐	☒	☐
Model configuration	☐	☐	☒	☐	☐
Training	☐	☒	☐	☐	☐
Evaluation	☐	☒	☐	☐	☐
Plotting/Diagnostics	☐	☐	☐	☒	☐

B3. Notes (issues, decisions, or gaps to revisit):

B5. Time spent per step (record for this use-case; minutes or start/end)

Step	Start (hh:mm)	End (hh:mm)	Minutes
Preprocessing	_____	_____	<u>226</u>
Model configuration	_____	_____	<u>23</u>
Training	_____	_____	<u>5</u>
Evaluation (metrics/baseline)	_____	_____	<u>5</u>
Plotting / Diagnostics	_____	_____	<u>52</u>
Debugging (errors, crashes, outputs)	_____	_____	<u>≥ 60</u>
Extra learning / searching / docs	_____	_____	<u>≥ 120</u>
Total			<u>≥ 491</u>

Notes: “Learning” includes reading docs, web searches, tutorials, setup notes, and guides. “Debugging” covers error fixing, crashes, failed runs, and investigating unexpected results.

B6. Comments (bugs, friction points, ideas):

Section E – IoT Use Case Template Reuse (Part 2)

Instructions: This section evaluates your experience reusing the provided IoT pipeline template (e.g., River Flow Forecasting) and adapting it for your own use case. Please answer honestly.

E1. Template Understanding

How clear was the IoT pipeline template (overview, scripts, execution order)?

☐ 1 Very unclear ☐ 2 Unclear ☐ 3 Neutral ☐ 4 Clear ☒ 5 Very clear

E2. Adaptation Effort

How easy was it to adapt the template to your chosen use case?

☐ 1 Very difficult ☐ 2 Difficult ☐ 3 Neutral ☒ 4 Easy ☐ 5 Very easy

E3. Reuse of Offline Training Code

How straightforward was it to reuse the provided offline training code with your dataset?

☐ 1 Not at all ☐ 2 A little ☐ 3 Neutral ☒ 4 Mostly ☐ 5 Very straightforward

Understanding the template: _____
E4. Time & Effort (minutes) Adapting scripts (sensors, gateway, analytics): 30 min
Running end-to-end pipeline: 30 min

E5. Problems & Debugging

Did you face errors while reusing the template? ☒ Yes ☐ No

If yes, briefly describe the main issues: Reshape function on analytics.py, in the first time I used reshape(1, -1) instead of reshape(1, lag * len(feature_cols)).

How difficult were they to fix?

☐ 1 Very easy ☒ 2 Easy ☐ 3 Neutral ☐ 4 Hard ☐ 5 Very hard

E6. Usefulness

How useful was the template for accelerating your implementation?

☐ 1 Not useful ☐ 2 Slightly useful ☐ 3 Neutral ☐ 4 Useful ☒ 5 Very useful

E7. Suggested Improvements

What changes would make the IoT template easier to reuse for future use cases?

I think the template is already very clear and easy to adapt.

Scale anchors (for consistency):

- Clarity / Ease: 1 = Very unclear/difficult, 2 = Unclear/difficult, 3 = Neutral, 4 = Clear/easy, 5 = Very clear/easy.
- Debugging difficulty: 1 = Very easy, 5 = Very hard.
- Usefulness: 1 = Not useful, 5 = Very useful.

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B1. Which steps did you complete? (check all that apply)

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Understanding the template:
E4. Time & Effort (minutes) Adapting scripts (sensors, gateway, analytics):
Running end-to-end pipeline:

30 min

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